PROBLEM CL21

Array of Hot Cylinders in a Gas Stream: Density, Force and Heat

Original statement · CL21.pdf

Periodicity | Drag | Heat release | Perfect gas

OFFICIAL PROBLEM STATEMENT

A linear array of parallel hot cylinders of radius R, whose centers are separated a distance $3R$, is placed perpendicular to a gas stream with velocity U_1, density ρ_1, and pressure p_1, as indicated in the figure of CL21.pdf. Sufficiently far downstream the gas properties recover uniform values, including a velocity U_2 and a pressure p_2. Assume that the effect of gravity is negligible. In terms of the quantities indicated in the figure, determine:

- The value of the density downstream from the array of cylinders.
- The force acting on each cylinder (per unit length).
- The heat transferred from each cylinder to the gas stream per unit time (per unit length). Express the result with use made of the ratio of specific heats γ.

NOTE: Explain clearly the simplifications and assumptions employed in the derivation.

Source: CL21.pdf (page 1, retake of Jun 13, 2014). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, planar (per unit length of cylinder); neglect $\bar{g}$; perfect gas, $h = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$; uniform states far upstream (1) and far downstream (2), where viscous stresses and conduction vanish.
- Infinite array: the flow is periodic in y with period $3R$ and symmetric about each cylinder axis, so the downstream velocity stays along $\bar{e}_x$ (no lift).
- Cylinders at rest (no work); they release heat to the gas.
- Data: $[\rho_1,\ U_1,\ p_1,\ U_2,\ p_2,\ R,\ \gamma]$. Unknowns: $[\rho_2,\ \bar{F},\ \dot{Q}]$ per unit length.

Symbol	Meaning	SI unit
U_1, ρ_1, p_1	upstream state	m/s, kg/m^3, Pa
U_2, ρ_2, p_2	downstream state	m/s, kg/m^3, Pa
R	cylinder radius (spacing $3R$)	m
$\bar{F}, \dot{Q}$	force and heat per unit length	N/m, W/m

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the array; $\bar{e}_x$ along the stream. Fixed control volume of height $3R$ around one cylinder (as in the official figure).
- Surfaces: Σ_1 (far upstream, $\bar{n} = -\bar{e}_x$), Σ_2 (far downstream, $\bar{n} = +\bar{e}_x$), Σ_{top} and Σ_{bot} (lines midway between cylinders, $\bar{n} = \pm\bar{e}_y$) and Σ_{cyl} (cylinder surface). By periodicity every flux through Σ_{top} equals minus the one through Σ_{bot}: they cancel each other out.

$$\boxed{\bar{F} = \int_{\Sigma_{cyl}} \left[p\,\bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

$$\boxed{\dot{Q} = -\int_{\Sigma_{cyl}} \bar{q} \cdot \bar{n}\, d\sigma}$$

PHASE 3

Balances, Surface by Surface

1. Mass (Notes Eq. 3.11)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\, dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho \left(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0} \right) \cdot \bar{n}\, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_1	$-\rho_1 U_1\, 3R$
Σ_2	$+\rho_2 U_2\, 3R$
$\Sigma_{top}+\Sigma_{bot}$	0 (cancel each other out)
Σ_{cyl}	0

$$\boxed{\rho_2 = \frac{U_1}{U_2}\,\rho_1} \tag{1}$$

2. Linear momentum (Notes Eq. 3.36)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c} p\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_1	$-\rho_1 U_1^2\, 3R\,\bar{e}_x$	$+p_1\, 3R\,\bar{e}_x$	0
Σ_2	$+\rho_2 U_2^2\, 3R\,\bar{e}_x$	$-p_2\, 3R\,\bar{e}_x$	0
$\Sigma_{top}+\Sigma_{bot}$	0	0	0 (cancel each other out)
Σ_{cyl}	0	$-\int_{\Sigma_{cyl}} p\,\bar{n}\,d\sigma$	$\int_{\Sigma_{cyl}}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed terms: $-\bar{F}$. With (1), $\rho_2 U_2^2 = \rho_1 U_1 U_2$.

$$\boxed{\frac{\bar{F}}{3R} = \left[p_1 - p_2 + \rho_1 U_1^2\left(1-\frac{U_2}{U_1}\right)\right]\bar{e}_x} \tag{2}$$

3. Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\left(e+\frac{v^2}{2}\right)dV}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\left(e+\frac{v^2}{2}\right)\left[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}\right]d\sigma=-\int_{\Sigma_c}p\,\bar{v}\cdot\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma+\underbrace{\int_{V_c}\rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect }\bar{g}}-\int_{\Sigma_c}\bar{q}\cdot\bar{n}\,d\sigma+\underbrace{\int_{V_c}Q_R\,dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int\rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int\bar{q}\cdot\bar{n}\,d\sigma$
Σ_1	$-\rho_1 U_1\,3R\left(e_1+\frac{U_1^2}{2}\right)$	$+p_1U_1\,3R$	0	0
Σ_2	$+\rho_2 U_2\,3R\left(e_2+\frac{U_2^2}{2}\right)$	$-p_2U_2\,3R$	0	0
$\Sigma_{top}+\Sigma_{bot}$	0	0	0	0 (cancel each other out)
Σ_{cyl}	0	0 ($\bar{v}=0$)	0	$-\int_{\Sigma_{cyl}}\bar{q}\cdot\bar{n}\,d\sigma$

Boxed term: $\dot{Q}$. With $h=\frac{\gamma}{\gamma-1}\frac{p}{\rho}$ and $\frac{p_2}{\rho_2}=\frac{p_2U_2}{\rho_1U_1}$ from (1):

$$\boxed{\dot{Q}=\rho_1U_1\,3R\left[\frac{\gamma}{\gamma-1}\left(\frac{p_2U_2}{\rho_1U_1}-\frac{p_1}{\rho_1}\right)+\frac{1}{2}\left(U_2^2-U_1^2\right)\right]}\qquad(3)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- Heating expands the gas: $\rho_2<\rho_1$ requires $U_2>U_1$ (1). The drag (2) combines the pressure drop across the array with the momentum change.
- Without heat ($\dot{Q}=0$) and without drag, (1) to (3) are satisfied by state 2 = state 1.
- Dimensions (SI, per unit length): $[(p+\rho U^2)R]=\mathrm{N/m}$; $[\rho U R\,p/\rho]=\mathrm{W/m}$.
- Numerical illustration (hypothetical data): $\rho_1=1.2$ kg/m^3, $U_1=30$ m/s, $p_1=101325$ Pa, $U_2=31$ m/s, $p_2=101000$ Pa, $R=0.01$ m, $\gamma=1.4$ give $\rho_2=1.1613$ kg/m^3, $F/L=8.67$ N/m and $\dot{Q}/L=9614.2$ W/m.
- Agreement with the official solution: CL21.pdf obtains (1), (2) and (3) (the handwritten correction in (3) replaces $1/\rho_2$ by $U_2/(\rho_1U_1)$, as here).